Response to Reviewers on "Fast and Flexible Inference for Spatial Extremes"

June 29, 2026

Response to Referee 1

The paper under view studies statistical max-stable and r-Pareto models for spatial extremes that allow for efficient inference via the so-called spectral likelihood. As already stated in the literature, the applicability of this method relies on the fact that the exponent measure of any vector $(Z(s_1), \dots, Z(s_D))$ has all its mass in the interior of $[0, \infty)^D$. Thus, the authors provide an equivalent criterion for this, based on conditional distributions of the spectral process. Furthermore, they propose two new classes of models extending/adapting the well-known Brown–Resnick and extremal-t processes to this setting leading to the skewed Brown–Resnick and truncated extremal-t models, respectively. The paper addresses an important and methodologically challenging question – efficient inference for high-dimensional extremes. It contains a number of novel contributions. My feeling, however, is that these novel elements are too limited to target a journal like JASA. More precisely:

- *Even though the equivalent criterion in terms of conditional probabilities is new, the fact that the exponent measure (and angular measure) does not put any mass on the boundary if the spectral process is strictly positive seems to be a direct consequence from other results in the literature, e.g., in the cited work by Dombry and co-authors such as Dombry and Ribatet (2015).*
- *The skewed Brown–Resnick process model is a straightforward extension of the multivariate Hüsler–Reiss model by Padoan (2011) and Beranger et al. (2019). Furthermore, to become a truly spatial model, the scale matrix Σ should be replaced by a (positive definite) scale function already in its definition.*
- *The definition of the truncation looks a bit strange. I guess that you mean*

$$(\tilde{Y}(s_1), \dots, \tilde{Y}(s_D)) \stackrel{d}{=} (Y(s_1), \dots, Y(s_D)) \mid \min_{i=1}^{D} Y(s_i) > 0.$$

  However, this is not a truly spatial model in the sense that it does not directly extend to a continuous spatial domain $\mathcal{S}$. In this case, an appropriate condition would be that $\inf_{s \in \mathcal{S}} Y(s) > 0$; however, this does not appear to lead to tractable finite-dimensional likelihoods.

Considering these reservations regarding the manuscript's primary methodological contributions, I believe that the results, as well as the model once it is fully spatial, merit publication; however, JASA T&M may not be the most suitable outlet.

Response:

Response to Referee 2

Major Comments

(i) Scope and title. *Recent work in spatial extremes has moved beyond asymptotic classes toward sub-asymptotic constructions such as random scale mixtures (e.g., Huser et al., 2017; Huser and Wadsworth, 2019) and max-infinitely-divisible models (e.g., Reich and Shaby, 2012; Bopp et al., 2021); see Huser and Wadsworth (2022) and Huser et al. (2024) for comprehensive reviews and comments. In fact, Huser et al. (2024) cautioned the community to move away from max-stable processes (or their corresponding r-Pareto processes) for modelling environmental data due to their stringent dependence properties. Given this context, the present title overstates generality. Please consider a more precise title such as "Fast spectral inference for max-stable and r-Pareto models".*

Response:

(ii) The max-stability assumption. *The reason that the sub-asymptotic models are preferred over max-stable processes is not only because we want to get the dependence class right, but also because we want to use a model that can accommodate weakening dependence structure as the level of extremity increases so that the extreme clusters in the model will appear more localised as they become more intense. For any max-stable process with unit Fréchet margins, the extremal coefficient of two locations with distance h becomes almost constant after a certain level of extremity:*

$$\theta_h(z) - \theta_h = O(1/z), \quad z \to \infty,$$

in which $\theta_h = 2 - \lim_{z\to\infty} \Pr\{Z(s_1) > z \mid Z(s_2) > z\}$. The corresponding r-Pareto also has this stability property. Although anisotropy in the proposed skewed Brown–Resnick may improve realism, this stability property remains and imposes a potentially unrealistic assumption on storm sizes for extreme rainfall or heatwave extents for extreme temperature. Both Huser and Wadsworth (2019) and Hazra et al. (2025) can exhibit weakening dependence strength as the quantile level increases. Furthermore, the skewed Brown–Resnick process and the truncated extremal-t process are asymptotically dependent everywhere over the large spatial domains considered in the applications of this paper; $\theta_h \to 2$ only if $h \to \infty$. This will inflate the risks associated with the simultaneous occurrence of extremes over space.

Response:

(iii) Model validity on large domains (need for testing). *Both proposed models are asymptotically dependent for any finite h on the large geographic domains analysed. Before fitting,*

the manuscript should test max-stability/max-domain suitability, e.g., via the space-time max-stability test of Koh et al. (2025) (or comparable diagnostics). In experience with SST fields (Hazra and Huser, 2021), max-stability is rejected even on subregions of 100 km, with rapid transition to asymptotic independence beyond modest lags. If a max-stable/r-Pareto model is used, restrict to subregions where the assumption is empirically defensible and say so explicitly. It might be justifiable to use a max-stable or r-Pareto process to model the precipitation in a smaller spatial domain such as central Florida, but the authors need to specifically show that.

Response:

(iv) Careless notations. *The authors use notations very frivolously, which causes immense confusion and makes the paper very difficult to read. For example, in Theorem 2, the inline notation is very dense and hard to parse. Specific issues include:*

- *The bold (for vectors) and roman (for scalars) fonts are inconsistently used.*
- *In $\overline{\Sigma} = \omega^{-1}\Sigma\omega$, it is unclear whether ω is a scalar or a matrix.*
- *The matrix $A_k = (e_1, \ldots, e_{k-1}, -\mathbf{1}, e_k, \ldots, e_{D-1})$: the role of e_k is not defined clearly.*
- *The subscript in $\Sigma_{\cdot k}$ is not defined.*
- *The superscripts in $x^{\circ\circ}_{-k}$ are unnecessarily clumsy.*
- *The notations w_k and I in Equation (15) are not defined.*
- *The function Φ° in Equation (17) is never defined in the main text.*
- *Under Equation (11), $\Pr(W \in [0, xr])$ makes no sense.*

There are many more notations that are not consistent throughout the text. The authors must carefully correct all the mistakes.

Response: We sincerely apologise for the notational inconsistencies. We have thoroughly revised the notation throughout the paper, ensuring consistent use of bold fonts for vectors and matrices, clearly defining all symbols at first use, and moving supplementary definitions into the main text where necessary. A full notation table has been added.

(v) Computational evidence and baselines. *"More efficient than state-of-the-art" (Line 14 of Page 30) needs quantification. Please report wall-clock times and memory for simulation and both data sets, with/without anisotropy, knots, etc. and comparisons to contemporary baselines such as neural Bayes estimators (Sainsbury-Dale et al., 2025). Moreover, the applications analyse 3,600 (Florida rainfall) and 1,043 (Red Sea SST) locations, whereas the simulation study uses only $D \in \{100, 225\}$; thus the simulation results do not demonstrate inferential performance at application scale.*

Response:

(vi) Are 4–5 knots for η adequate on large, heterogeneous domains (Florida; Red Sea)? Related work uses richer local bases (e.g., 36 in Hazra et al., 2025 for a much smaller region). Please justify knot counts and provide sensitivity checks.

Response:

Minor Comments

(i) Page 5 — If $C^+(S)$ is a function space and $B \subset C^+(S)$, how do you evaluate $\Pr(X/u \in B)$ if X is a real-valued spatial process?

Response: We have simplified the notation, and it should be clear now.

(ii) Page 6 — You wrote "$\frac{(\tilde{W}(s))^{\nu+}}{a(s)}$, where $\tilde{W}_+(s) = \max\{\tilde{W}(s), 0\}$". Please be consistent.

Response: We have corrected the notation to be consistent throughout.

(iii) Equation (7) — Should "A" be "A_u"?

Response: Yes, this was a typo. We have corrected it.

(iv) Page 8 — Should "$r(x) = \|x\|_1$" be "$r_0(x) = \|x\|_1$"?

Response: No, this is correct as written.

(v) Page 15 — The constraint "$\eta 1 = 0$" came out of nowhere. Why do you need it?

Response: Commented.

Response to Referee 3

I worry that the title is too general and may miss the specificities of the paper. In particular, the restriction of the new models developed to the asymptotic dependent case does rule out a substantial fraction of spatial extremes.

Response: We agree and have revised the title to better reflect the scope of the paper (see also our response to Referee 2, Major Comment (i)).

Pg 2, l51: 'do not provide any statistical guarantees'. Rödder et al. (2025) seem to suggest there is some justification for these neural Bayes estimators.

Response:

Pg 2, l7: 'Vecchia approximation'. More related to your paper on r-Pareto processes and non-stationarity, Koh et al. (2025) fit an r-Pareto model using the gradient score and the Vecchia approximation in an ML algorithm.

Response:

Pg 5, l44: 'The fact that κ can put mass on $\partial\Omega$ hinders the convergence of X, introducing large bias in estimation'. I find this interesting, and I am not currently aware of an obvious reference for this. In any case, I think it would be nice to show a simulation study showing this bias.

Response:

Some typos: pg 18, l41: 'the method'. pg 24, l52: 'is is'.

Response: We have corrected these typos throughout the manuscript.

Section 4.3: I found your notation of η a little confusing, because it is not always clear if it is a vector or a scalar. For example in pg. 14, l52: $\eta = 0$, but above in l31: ξ has a one-to-one relationship with η, but then $\xi = 0$, which is a scalar?

Response: We agree the notation was confusing. We have revised Section 4.3 to consistently use $\boldsymbol{\eta}$ for the vector and η_k for its components, and clarified the relationship between ξ and η_k.

Section 5.2: How robust is your estimation procedure to not knowing the true centre of the kernels? In practice I can imagine that knowing the true centres is difficult. Is the η parameter vector difficult to estimate?

Response:

Pg 19, l27: What is the intuition for choosing this value of $u = 30 \times D$? I understand this should be large but this feels rather arbitrary.

Response:

Fig 3: Isn't the bias you see from the spectral likelihood concerning? Also shouldn't the existing theory say that the pairwise likelihood approach would give unbiased estimates for all the parameters (including b_1 and b_2)?

Response:

Pg 22, l44: 'There is no bias (or approximation) for r-Pareto processes'. Isn't there the approximation when you use the likelihood of the r-Pareto process with L_1 norm as risk functional to make inference for a different risk functional?

Response:

I am generally a little confused with how exactly you are using the trick of choosing $u > D^{1-1/p}$ to go from the L_1 to L_p. This should be spelled out a bit more. Also what does this mean exactly in practice? Do you have to choose a higher u for higher p (L_p) risk functionals? For the data application, page 24, l38: you choose the 99.95% for both risk functionals L_1 and L_∞ as that is where the estimate shape parameter becomes stable, but how is this linked to the trick of looking at the L_1 Pareto process? This is also related to pg 30, l3: 'can be bypassed by simply considering the likelihood of an r-Pareto process with L_1 risk functional'.

Response:

Relatedly, what threshold did you choose in Fig 3 for the L_3 r-Pareto simulation study?

Response:

Pg 27, l12: You chose $u = 10 \times D$ here. What exactly are the guidelines for choosing u more generally? Both in the r-Pareto and max-stable cases? Also, here you have 9 observations.

And if I am not mistaken, for up to 9 parameters? How is this still feasible?

Response:

Is there existing theory on using the spectral likelihood's asymptotic variance instead of using the jackknife for uncertainty quantification? This could be nice so you can reduce computation for the uncertainty quantification. But I guess you are quite far from asymptotics if you only use as little as 9 observations.

Response:

References

Dombry, C. and Ribatet, M. (2015) Functional regular variations, Pareto processes and peaks over threshold. *Statistics and Its Interface* **8**, 9–17.